

The Keynesian Cross, Fiscal Policy, and the IS Curve

Session 5 – Introduction to Macroeconomics

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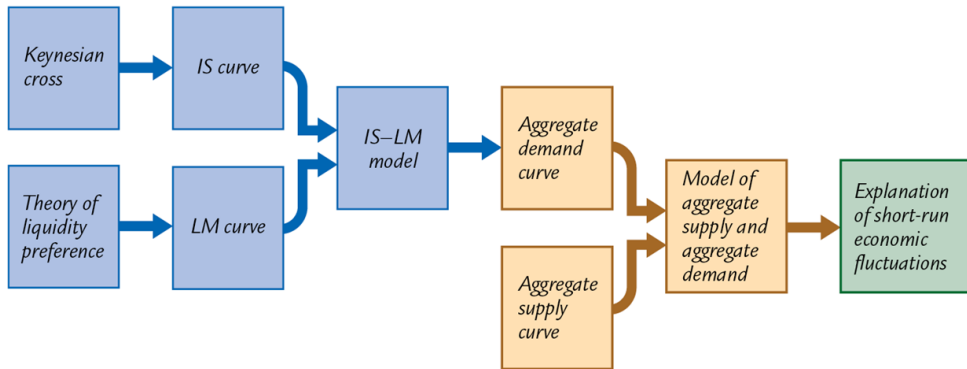
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- ▶ Our framework is **AS–AD**. Last time, we focused on supply:
 - ▶ What does **LRAS** look like? Vertical at $Y = \bar{Y}$.
 - ▶ What does **SRAS** look like? Horizontal at a fixed price level.

The Overall Plan



Mankiw, *Macroeconomics*, 11e, © 2022 Worth Publishers

- **Last time:** aggregate supply, taking AD as given.
- **Today:** Keynesian cross → IS curve, the goods-market building block of AD.

Source: Mankiw, *Macroeconomics*, 11th ed. (2022), Worth Publishers.

Session 5: Outline

1. **The Keynesian Cross**

Planned expenditure, equilibrium, and a numerical example.

2. **Fiscal Policy and Multipliers**

Government purchases, taxes, and the spending feedback.

3. **Extensions**

Autonomous consumption, income taxes, and trade.

4. **Output Gaps and Fiscal Policy**

Potential output, recessionary gaps, and inflationary pressure.

5. **Building the IS Curve**

Investment, interest rates, and goods-market equilibrium.

Reading: Mankiw, *Macroeconomics*, 12th ed., Chapter 13: the Keynesian cross and the IS curve.

1. The Keynesian Cross

How do spending plans determine short-run output?

The Keynesian Cross

- ▶ A model of the relationship between **expenditure and income**, used to determine equilibrium output.
- ▶ Three components:
 1. **Planned expenditure**: what the economy plans to spend.
 2. **Actual expenditure**: the value of output produced.
 3. **Equilibrium**: planned expenditure equals output.

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- ▶ Three components:
 1. **Planned expenditure**: what the economy plans to spend.
 2. **Actual expenditure**: the value of output produced.
 3. **Equilibrium**: planned expenditure equals output.
- ▶ Our starting assumptions:
 - ▶ Fixed prices; firms adjust production to demand.
 - ▶ A closed economy, $NX = 0$; $I = \bar{I}$, $G = \bar{G}$, $T = \bar{T}$.
- ▶ Output Y is **endogenous**; it need not equal $\bar{Y}$.

Planned Expenditure

- ▶ **Planned expenditure**, PE , is what households, firms, and government intend to spend on domestic output:

$$PE = C + \bar{I} + \bar{G}.$$

- ▶ Households consume part of their disposable income, $Y - \bar{T}$. Assume $C = c \times (Y - \bar{T})$, where $0 < c < 1$.
- ▶ **What is the marginal propensity to consume (MPC)?**

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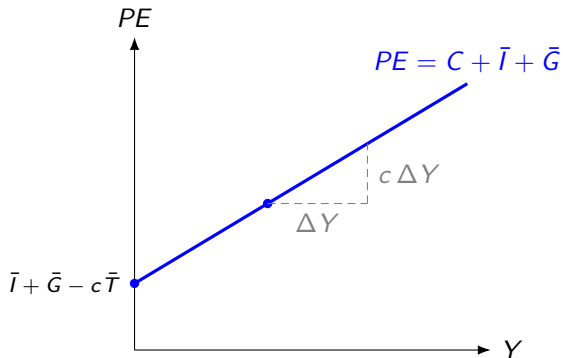
$$PE = c \times (Y - \bar{T}) + \bar{I} + \bar{G} = cY + \underbrace{\bar{I} + \bar{G} - c\bar{T}}_{\text{independent of } Y}.$$

Why Does Planned Expenditure Rise with Income?

- ▶ Higher income raises consumption and therefore planned expenditure.
- ▶ With fixed taxes and investment:

$$\Delta PE = c \Delta Y.$$

- ▶ The slope is $c = \text{MPC}$: positive, but smaller than one.



Actual Spending Is Not Always Planned Spending

- ▶ National accounts record **actual expenditure** as equal to output, Y .
- ▶ If firms sell less than expected, unsold goods enter inventories. These count as investment, even when **unplanned**.

$$\underbrace{Y}_{\text{actual expenditure}} = \underbrace{C + \bar{I} + \bar{G}}_{\text{planned expenditure}} + \underbrace{I^u}_{\text{unplanned inventory investment}}.$$

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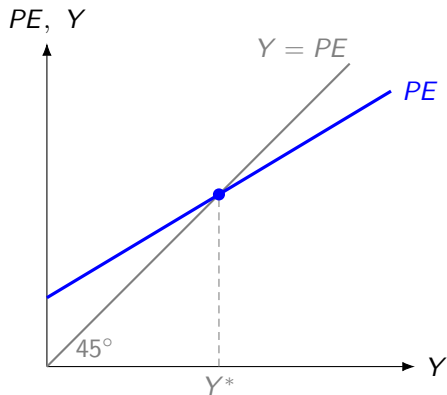
- ▶ **Goods-market equilibrium:** spending plans are fulfilled, so

$$I^u = 0 \quad \Longleftrightarrow \quad Y = PE.$$

- ▶ No *unplanned* inventory change does not mean no inventories or no planned inventory investment.

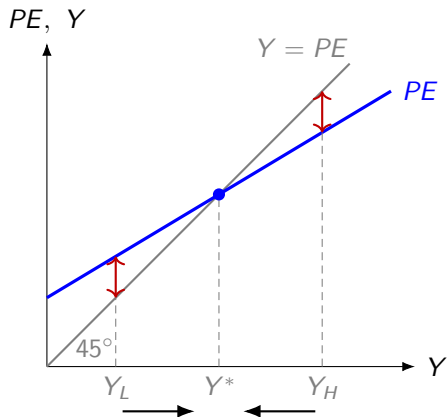
The Keynesian Cross: Equilibrium and Adjustment

- ▶ The 45° line shows where expenditure equals output.
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- ▶ At the intersection, $Y^* = PE$: firms have no reason to change production.
- ▶ If $Y < Y^*$: $PE > Y$. Inventories fall unexpectedly; firms **raise output**.
- ▶ If $Y > Y^*$: $PE < Y$. Inventories accumulate unexpectedly; firms **cut output**.



A Numerical Example: Find Equilibrium Output

$$\bar{G} = 10, \quad \bar{T} = 5, \quad \bar{I} = 14, \quad C = 0.8(Y - \bar{T}).$$

What is the equilibrium level of Y ?

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$$PE = 0.8(Y - 5) + 14 + 10 = 0.8Y + 20,$$

$$Y = PE \Rightarrow Y = 0.8Y + 20,$$

$$0.2Y = 20 \Rightarrow Y^* = 100.$$

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Check: $C = 0.8(100 - 5) = 76$, so $C + I + G = 76 + 14 + 10 = 100$.

All amounts are in the same real units per period. Output equals *planned* spending at $Y = 100$.

What If Firms Produce Too Much or Too Little?

Keep the same example: $PE = 0.8Y + 20$.

What happens if firms initially produce 80? What if they produce 120?

Output Y	Planned spending PE	$I^u = Y - PE$	Firms' response
80			
100			
120			

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Output Y	Planned spending PE	$I^u = Y - PE$	Firms' response
80	84	-4	Raise output
100	100	0	No adjustment
120	116	4	Cut output

The accounting identity still holds away from equilibrium: **unplanned inventory investment makes up the difference.**

2. Fiscal Policy and Multipliers

Why can a change in spending have a larger effect on output?

What Is Fiscal Policy?

- ▶ **Fiscal policy** is the government's choice of purchases, taxes, and transfers.
 - ▶ G : purchases of goods and services, such as schools and roads.
 - ▶ T : **net taxes**, taxes paid minus transfers received by households.
 - ▶ Transfers affect disposable income; they are not purchases in G .

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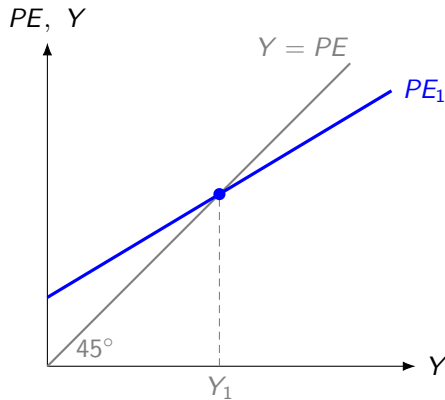
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- ▶ **Expansionary** fiscal policy raises demand: higher G or lower T .
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- ▶ Bars mean G and T are **exogenous within an equilibrium**. We can compare equilibria under different policy choices.

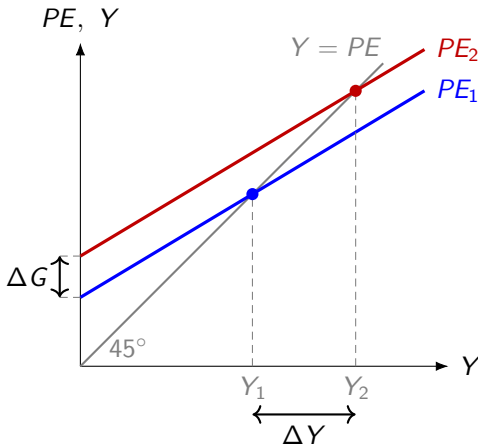
An Increase in Government Purchases

- Suppose G rises. **What happens to planned expenditure and equilibrium output?**



An Increase in Government Purchases

- ▶ Suppose G rises. **What happens to planned expenditure and equilibrium output?**
- ▶ The PE line shifts up by ΔG at every income level.
- ▶ Equilibrium output rises from Y_1 to Y_2 .
- ▶ **Why is $\Delta Y > \Delta G$?**



Why Does Output Rise by More Than Government Purchases?

- ▶ The government buys more goods and services. Firms increase production, and households earn more income.
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 - ▶ Their extra spending becomes income for other households.
- ▶ Those households also spend part of their extra income. The process repeats:

$$G \uparrow \Rightarrow Y \uparrow \Rightarrow C \uparrow \Rightarrow Y \uparrow \Rightarrow C \uparrow \Rightarrow \dots$$

- ▶ The initial purchase generates **further rounds of spending**. How large is the total effect?

Quantifying the Multiplier

MPC reminder: households consume a fraction c of additional disposable income and save the remaining fraction $1 - c$.

In our example, $c = 0.8$. Suppose government purchases rise by 2.

Round	Additional spending and income	Amount
Initial purchase	ΔG	2

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Round	Additional spending and income	Amount
Initial purchase	ΔG	2
First consumption response	$c\Delta G$	1.6
Next response	$c^2\Delta G$	1.28
And so on	$c^3\Delta G + c^4\Delta G + \dots$	$\dots$

$$\Delta Y = (1 + c + c^2 + \dots)\Delta G = \frac{1}{1 - c}\Delta G = 10.$$

Each round is smaller, but the **total** is larger than the initial purchase.

Deriving the Government-Purchases Multiplier

Compare two equilibria, keeping taxes and planned investment fixed:

$$Y = C + \bar{I} + G,$$
$$\Delta Y = \Delta C + \Delta G = c\Delta Y + \Delta G.$$

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$$(1 - c)\Delta Y = \Delta G \quad \Rightarrow \quad k_G \equiv \frac{\Delta Y}{\Delta G} = \frac{1}{1 - c}.$$

- ▶ k_G is the change in equilibrium output per unit increase in G .
- ▶ With $c = 0.8$, $k_G = 5$: G rises from 10 to 12, and Y from 100 to 110.
- ▶ This result assumes **fixed prices, fixed investment, fixed taxes, and no imports**. It is a model result, not a universal empirical constant.

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- ▶ With $c = 0.8$, $k_T = -4$. A tax cut of 2 ($\Delta T = -2$) raises Y by 8.
- ▶ Unlike a government purchase, part of a tax cut is saved *before the first spending round*: $|k_T| = c k_G < k_G$.

Compare the Multipliers

Change	Initial shift in PE	Multiplier	If $c = 0.8$
Government purchases ΔG	ΔG	$1/(1 - c)$	5
Net taxes ΔT	$-c\Delta T$	$-c/(1 - c)$	-4
Planned investment $\Delta \bar{I}$	$\Delta \bar{I}$	$1/(1 - c)$	5

Quick application: from the original equilibrium $Y = 100$, firms reduce planned investment by 3. What happens?

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Quick application: from the original equilibrium $Y = 100$, firms reduce planned investment by 3. What happens?

$$\Delta Y = 5(-3) = -15 \quad \Rightarrow \quad Y_{\text{new}} = 85.$$

The same spending feedback can **amplify a fall in demand**.

3. Extensions

Autonomous consumption, income taxes, and trade

Add: Autonomous Consumption and Income Taxes

- ▶ Add **autonomous consumption**, $\bar{C}$, independent of current disposable income:
 $C = \bar{C} + c(Y - T)$.
 - ▶ Example: a household with no current income still buys food, using past savings or borrowing.

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- ▶ The MPC out of *disposable* income is still c , but an extra unit of *total* income raises consumption by only $c(1 - t)$.

Equilibrium with Proportional Taxes

From this equilibrium condition, how do we solve for Y ?

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Extend our example: $\bar{C} = 12$, $c = 0.8$, $t = 0.2$, $\bar{I} = 14$, $\bar{G} = 10$.

$$PE = 12 + 0.64Y + 14 + 10 = 36 + 0.64Y, \quad Y^* = \frac{36}{0.36} = 100.$$

What is the government-purchases multiplier now?

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What is the government-purchases multiplier now?

$$k_G = \frac{1}{1 - c(1 - t)} = \frac{1}{0.36} \simeq 2.78.$$

- ▶ The multiplier is smaller than 5: taxes dampen the spending feedback.
- ▶ This is an **automatic stabilizer**: tax payments respond to income even without a new policy decision.
- ▶ Changing $\bar{C}$ shifts PE ; changing t changes its slope.

Two Additions: Intercept, Slope, and Multiplier

To isolate the effects, start from $T = 0$. Keep $c = 0.8$, $\bar{I} = 14$, and $\bar{G} = 10$ fixed.

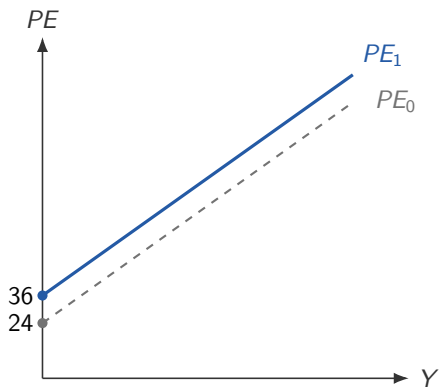
Baseline: $PE_0 = 24 + 0.8Y$.

1. **Add autonomous consumption:**

$$\bar{C} = 12.$$

$$PE_1 = 36 + 0.8Y.$$

The intercept rises from 24 to 36; the slope stays at 0.8. The multiplier stays at $k_G = 5$.



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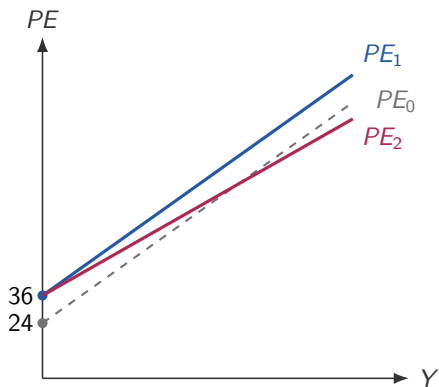
$$PE_1 = 36 + 0.8Y.$$

The intercept rises from 24 to 36; the slope stays at 0.8. The multiplier stays at $k_G = 5$.

2. **Add proportional taxes:** $T = 0.2Y$.

$$PE_2 = 36 + 0.64Y.$$

The intercept stays at 36; the slope falls to 0.64. Now $k_G = 1/(1 - 0.64) \simeq 2.78$.



Add: Exports and Imports (Open Economy)

- ▶ Exports X add foreign demand for domestic production.
- ▶ Imports M are subtracted because they are included in spending but are **not domestic output**.

First take exports and imports as given: $X = \bar{X}$, $M = \bar{M}$.

How do we solve for equilibrium output now?

$$Y = \bar{C} + c(1 - t)Y + \bar{I} + \bar{G} + \bar{X} - \bar{M}.$$

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$$\Rightarrow Y^* = \frac{\bar{C} + \bar{I} + \bar{G} + \bar{X} - \bar{M}}{1 - c(1 - t)}.$$

- ▶ Higher exports shift PE up; higher imports shift it down.
- ▶ **Fixed** imports change the intercept, not the multiplier. Next, we let imports respond to income.

What If Imports Rise with Income?

Let $M = \mu Y$, where $\mu \geq 0$ is the **marginal propensity to import**. Keep $X = \bar{X}$.

$$PE = \bar{C} + c(1 - t)Y + \bar{I} + \bar{G} + \bar{X} - \mu Y.$$

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$$PE = \bar{C} + c(1 - t)Y + \bar{I} + \bar{G} + \bar{X} - \mu Y.$$

$$[1 - c(1 - t) + \mu]Y = \bar{C} + \bar{I} + \bar{G} + \bar{X},$$

$$Y^* = \frac{\bar{C} + \bar{I} + \bar{G} + \bar{X}}{1 - c(1 - t) + \mu},$$

$$k_G = \frac{1}{1 - c(1 - t) + \mu}.$$

- ▶ Some additional spending goes to imports, not domestic production.
- ▶ Higher μ means a smaller domestic spending feedback and multiplier.

Imports Reduce the Domestic Multiplier

Hold $c = 0.8$ and $t = 0.4$ fixed; change the propensity to import:

μ	Slope of PE : $c(1 - t) - \mu$	$k_G = 1/[1 - c(1 - t) + \mu]$
0	0.48	1.92
0.2	0.28	1.39
0.4	0.08	1.09

Why Does the Multiplier Become Smaller?

Income taxes lower the multiplier; income-dependent imports lower it further.

Why? What happens to the next round of spending?

Why Does the Multiplier Become Smaller?

Income taxes lower the multiplier; income-dependent imports lower it further.

Why? What happens to the next round of spending?

- ▶ Less extra income returns as spending on **domestic output**:
 - ▶ Taxes reduce disposable income (with G held fixed).
 - ▶ Saving is not spent on consumption.
 - ▶ Imports generate income abroad, not domestic production.
- ▶ The stronger the withdrawals into saving, taxes, and imports, the smaller the domestic multiplier.
- ▶ Investment and prices are still fixed; interest-rate and price responses can change the multiplier.

Multiplier Summary: What Do We Assume?

Government-purchases multiplier: $k_G = \Delta Y / \Delta G$.

Model assumptions	k_G
Closed economy; fixed net taxes	$1/(1 - c)$
Add autonomous consumption; still fixed taxes	$1/(1 - c)$
Proportional taxes, $T = tY$	$1/[1 - c(1 - t)]$
Add fixed exports and imports	$1/[1 - c(1 - t)]$
Let imports respond to income, $M = \mu Y$	$1/[1 - c(1 - t) + \mu]$

All rows hold prices and planned investment fixed. Autonomous spending shifts PE ; income-dependent taxes and imports change its slope.

With fixed taxes in a closed economy: $k_T = -c/(1 - c)$ and $k_I = 1/(1 - c)$.

4. Output Gaps and Fiscal Policy

Can demand be too low or too high?

Potential Output and Inflationary Pressure

- ▶ **Potential output**, $\bar{Y}$, is the economy's sustainable output at normal resource utilization: its natural or full-employment level.
 - ▶ It depends on capital, labor, and technology.
 - ▶ Full employment does not mean zero unemployment.

Potential Output and Inflationary Pressure

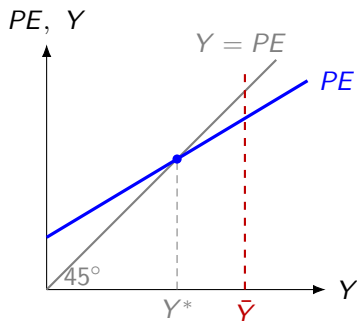
- ▶ **Potential output**, $\bar{Y}$, is the economy's sustainable output at normal resource utilization: its natural or full-employment level.
 - ▶ It depends on capital, labor, and technology.
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- ▶ It is **not a physical maximum**. Firms can temporarily use overtime and run equipment more intensively to produce above $\bar{Y}$.

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- ▶ It is **not a physical maximum**. Firms can temporarily use overtime and run equipment more intensively to produce above $\bar{Y}$.
- ▶ Strong demand can then create **inflationary pressure**: labor and other inputs become harder to obtain, encouraging wage and price increases.
- ▶ The Keynesian cross holds prices fixed. It identifies an *output gap*, $Y^* - \bar{Y}$, but does not predict an inflation rate.

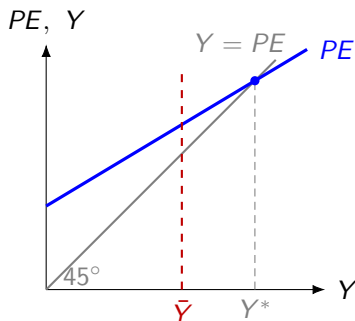
Equilibrium Need Not Be at Potential Output

Recessionary gap: $Y^* < \bar{Y}$



Weak demand; unemployment above its natural rate. Expansionary fiscal policy can raise output.

Inflationary gap: $Y^* > \bar{Y}$



Demand exceeds sustainable output. Contractionary fiscal policy can reduce pressure on resources.

Closing a Recessionary Gap: How Much Fiscal Policy?

Return to the **baseline closed economy** with fixed taxes and $c = 0.8$:

$$Y^* = 100, \quad \bar{Y} = 120, \quad k_G = 5, \quad k_T = -4.$$

The output shortfall is 20. Must government purchases rise by 20?

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Alternatively, with government purchases unchanged:

$$\Delta T = \frac{\bar{Y} - Y^*}{k_T} = \frac{20}{-4} = -5.$$

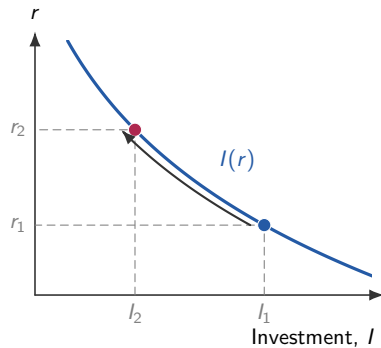
The required **initial spending change** is smaller than the output gap because subsequent consumption amplifies it. These are **alternative**, not simultaneous, policies.

5. Building the IS Curve

Let investment depend on the interest rate

Bring Back the Investment Function

- ▶ Return to the **closed economy with fixed taxes**.
- ▶ Replace $I = \bar{I}$ with $I = I(r)$, where r is the **real interest rate**.
- ▶ Higher r raises financing costs and the opportunity cost of investing: fewer projects are profitable.
- ▶ Thus $r \uparrow \Rightarrow I \downarrow \Rightarrow PE \downarrow \Rightarrow Y \downarrow$.

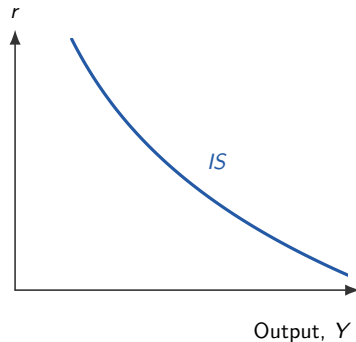


Constructing the IS Curve: Definition

IS plots all combinations of r and Y that give **goods-market equilibrium**:

$$Y = C(Y - \bar{T}) + I(r) + \bar{G}.$$

- ▶ Combine the **Keynesian cross** and the **investment demand function**.
- ▶ Lower interest rates stimulate investment and spending, giving higher equilibrium income.

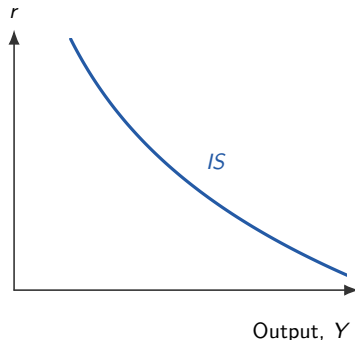


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- ▶ Combine the **Keynesian cross** and the **investment demand function**.
- ▶ Lower interest rates stimulate investment and spending, giving higher equilibrium income.
- ▶ This downward-sloping *demand-side relationship* holds G , T , and prices fixed. It is not yet AD, which relates P and Y .



Goods-Market Equilibrium: Why Call It IS?

We have defined IS using the goods market. **Why should its name refer to investment and saving?** Recall Session 2.

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$$Y = C + I + G \quad \Longleftrightarrow \quad \underbrace{Y - C - G}_{\text{national saving } S} = I.$$

$$S = \underbrace{Y - T - C}_{\text{private saving } S_p} + \underbrace{T - G}_{\text{public saving}}.$$

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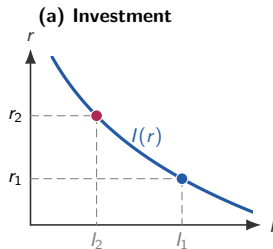
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- ▶ Each point on IS satisfies **national saving = planned investment**.
- ▶ Here income adjusts too: unlike the long-run model, Y is not fixed at $\bar{Y}$.

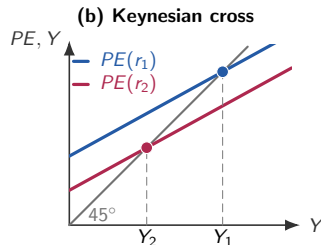
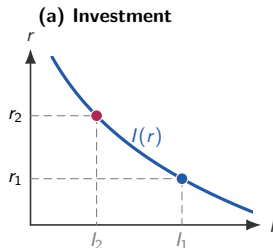
Constructing the IS Curve

1. **Panel (a):** raise r from r_1 to r_2 . Planned investment falls.



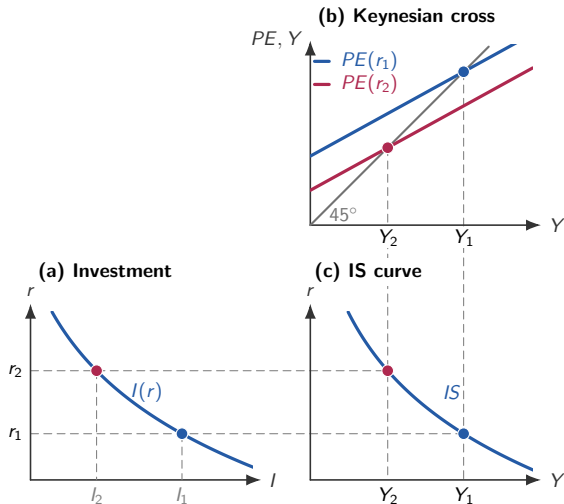
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Constructing the IS Curve

1. **Panel (a):** raise r from r_1 to r_2 . Planned investment falls.
2. **Panel (b):** lower investment shifts PE down. Equilibrium income falls from Y_1 to Y_2 .
3. **Panel (c):** plot the two goods-market equilibria. Higher r goes with lower Y : IS slopes down.



A Numerical IS Curve

For this numerical example, take $C = 0.8(Y - 5)$, $\bar{G} = 10$, and **linear investment** $I(r) = 24 - 2r$. Here r is measured in percentage points (so $r = 5$ means 5%).

A Numerical IS Curve

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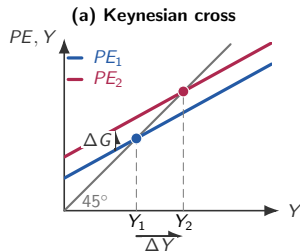
$$Y = 0.8(Y - 5) + (24 - 2r) + 10,$$
$$0.2Y = 30 - 2r \quad \Rightarrow \quad Y = 150 - 10r.$$

Real interest rate r	Planned investment I	Equilibrium Y
6%	12	90
5%	14	100
4%	16	110

A one-percentage-point fall in r raises investment by 2 and output by $5 \times 2 = 10$: this is a **movement along IS**.

Fiscal Policy Shifts the IS Curve

- **Panel (a):** at a given r , higher G shifts planned spending up. Equilibrium output rises.

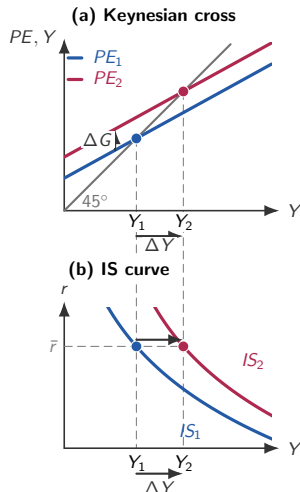


Fiscal Policy Shifts the IS Curve

- ▶ **Panel (a):** at a given r , higher G shifts planned spending up. Equilibrium output rises.
- ▶ **Panel (b):** this holds at *every* interest rate, so IS shifts **right**:

$$\Delta Y|_r = \frac{\Delta G}{1 - c}.$$

- ▶ In our numerical example, $c = 0.8$:
 $\Delta G = 2$ shifts IS horizontally by 10.
- ▶ This is the effect *holding r fixed*. Adding the money market will tell us how r responds.



What If Taxes Rise?

Suppose net taxes rise by 2, with $c = 0.8$. Does IS move or shift? In which direction?

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- ▶ At every interest rate, equilibrium output is lower: **IS shifts left.**

$$\Delta Y|_r = -\frac{c}{1-c} \Delta T = -4 \times 2 = -8.$$

- ▶ A tax cut has the opposite effect: it shifts IS right.

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- ▶ A tax cut has the opposite effect: it shifts IS right.

Keep the distinction clear:

- ▶ A change in r : a *movement along* IS.
- ▶ A change in G , T , or autonomous spending: a *shift of* IS.

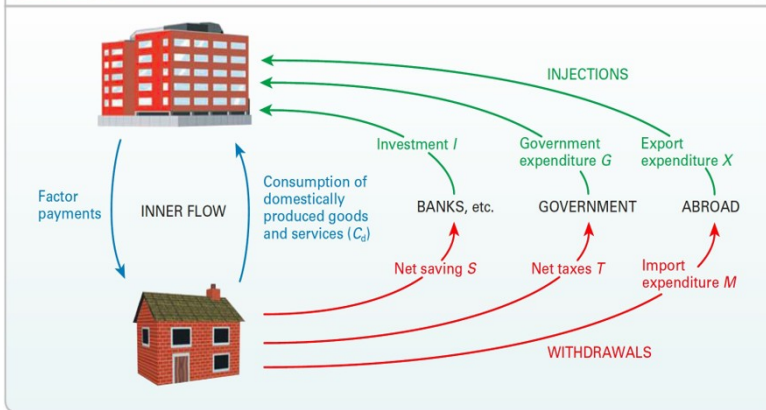
What to Remember

- ▶ **Actual spending equals output by accounting; planned spending equals output only in equilibrium.** Unplanned inventories explain the difference.
- ▶ **Spending creates further spending.** The multiplier depends on how much extra income returns as demand for domestic output.
- ▶ **Equilibrium need not mean full employment.** Demand can leave output below or above its sustainable level.
- ▶ **An interest-rate change moves us along IS; a fiscal-policy change shifts IS.** IS alone does not select one (Y, r) pair.

Next session: money and its supply. We then study the money market and LM, combining IS and LM to build aggregate demand.

Refresher: The Circular Flow

Figure 15.8 The circular flow of national income and expenditure



Withdrawals: saving, net taxes, and imports.

Injections: investment, government purchases, and exports.

In equilibrium:

$$S_p + T + M = I + G + X.$$

In this diagram, S denotes private saving (S_p in our notation), not national saving.

Source: Sloman, Garratt and Guest, *Economics*, 10th ed. (2018), Fig. 15.8.

[◀ Back to multipliers](#)

Practice: Government Purchases to Reach a Target

Target output is $Y = 1000$. In each case, $C = 200 + 0.7(Y - T)$ and $I = 100$.

Find the required G in each economy. Each row is a separate case.

Case	T at target	C at target	NX	Required G
Fixed $T = 100$; closed			0	
$T = 0.2Y$; closed			0	
$T = 0.2Y$; $X = 80$, $M = 0.16Y$				

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Case	T at target	C at target	NX	Required G
Fixed $T = 100$; closed	100	830	0	70
$T = 0.2Y$; closed			0	
$T = 0.2Y$; $X = 80$, $M = 0.16Y$				

Use $G = Y - C - I - NX$.

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Case	T at target	C at target	NX	Required G
Fixed $T = 100$; closed	100	830	0	70
$T = 0.2Y$; closed	200	760	0	140
$T = 0.2Y$; $X = 80$, $M = 0.16Y$				

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$T = 0.2Y$; closed	200	760	0	140
$T = 0.2Y$; $X = 80$, $M = 0.16Y$	200	760	-80	220

Use $G = Y - C - I - NX$.

At the same target output, higher taxes or greater net imports reduce other demand for domestic output, so more government purchases are required.

Practice: Is the Slope Always the MPC?

Suppose the consumption–income line is

$$C = 50 + 0.9Y, \quad \text{and} \quad T = 0.01Y.$$

What are autonomous consumption and the MPC out of disposable income?

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$$C = 50 + 0.9Y, \quad \text{and} \quad T = 0.01Y.$$

What are autonomous consumption and the MPC out of disposable income?

$$C = \bar{C} + c(Y - T) = \bar{C} + c(1 - t)Y,$$

$$\bar{C} = 50, \quad c(1 - 0.01) = 0.9,$$

$$c = \frac{0.9}{0.99} = \frac{10}{11} \simeq 0.909.$$

The distinction: 0.9 is the response to total income; 0.909 is the response to disposable income. They differ because taxes rise with income.